

3.9 Mathematical Requirements

In order to be able to develop the knowledge, understanding and skills in the specification, candidates need to have been taught, and to have acquired competence in, the areas of mathematics set out below. Material relevant to the second part of the A level (A2) only is given in bold type.

	<i>Candidates should be able to</i>
Arithmetic and computation	• recognise and use expressions in decimal and standard form • use ratios, fractions and percentages • make estimates of the results of calculations (without using a calculator) • understand the symbols > and < • use calculators to find and use mean, standard deviations and x^n, $1/x$, $\sqrt{x}$
Handling data	• use an appropriate number of significant figures • find arithmetic means • construct and interpret frequency tables, bar charts and histograms • understand the principles of sampling as applied to biological data • distinguish between chance and probability and understand the importance of chance and probability when interpreting data • understand the terms mean, median and mode and standard deviation • use a scatter diagram to identify positive and negative correlation between two variables • select and use a simple statistical test (see Teacher Resource Bank for further guidance) • Candidates are not required to recall statistical formulae but will be provided with an appropriate data sheet when necessary.
Algebra	• change the subject of an equation • substitute numerical values into algebraic equations using appropriate units for physical quantities • understand the use of logarithms in relation to quantities that range over several orders of magnitude.
Graphs	• translate information between graphical and numerical forms • plot two variables from experimental or other data using appropriate Institute of Biology conventions • calculate rate of change from a graph showing a linear relationship • draw and use the slope of a tangent to a curve as a measure of rate of change.
Geometry	• visualise three dimensional forms from two dimensional representations of three dimensional objects • calculate circumferences and areas of circles, surface areas and volumes of regular blocks and cylinders when provided with appropriate formulae.